

# Assessing a CompTIA Security+ Study Note for Credential and Access-Control Mastery

A Key-Displacement Study of a 25-Item Examination Protocol

[Author Name]

[Affiliation]

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## Abstract

Personal knowledge-management artifacts accumulate faster than they are validated, and a reader who trusts a study note inherits its defects silently. This paper treats one such artifact as an object of assessment rather than as an authority. A case-insensitive recursive search of the author's Obsidian vault localized exactly one candidate: a CompTIA Security+ textbook by Mark Ciampa, retained both as a source PDF and as a directory of fifteen extracted per-module notes. Scope was narrowed to Module 7, *Identity and Access Management* (SY0-701), which was read in full at 471 lines and is organized around three objectives: authentication credential types, authentication best practices, and access control schemes. A line-level audit exposed three defects in the note, including an internal contradiction on password expiry and a review item presenting two textually identical answer options. A 25-item examination protocol was then administered, each item carrying a nominal value of four points, under a requirement that every answer be scored and carry a line-level citation into the source. The supplied answer key failed an internal-consistency check: in roughly three quarters of its entries the printed option letter contradicted the printed explanatory rationale, and several entries were topically unrelated to any supplied item. Subject-matter re-mapping established that the rationales were reliable while the letters had been displaced; reissuing the same items with intra-question option reordering rendered the key internally consistent throughout, confirming displacement without resolving its cause. Rescoring against the aligned key yielded three accuracy figures under two weighting schemes: 72% under strict all-or-nothing scoring, 81% under selection-weighted scoring of multi-select items, and 86% under strict scoring with the four items judged defective or disputed excluded. Decomposition revealed marked heterogeneity: the five items newly introduced by the reissued set were answered without exception, while the twenty carried-over items yielded thirteen correct, with failure concentrated where the source note is silent on the distinction under test.

**Keywords:** study-material assessment; assessment validity; answer-key misalignment; access control; authentication; knowledge management.

**Scope note.** This is a single-artifact, single-analyst study-material assessment. No external literature was consulted. All references are internal artifacts. Quantitative claims are derived arithmetic from the session record; none have been extrapolated beyond the population actually observed.

# 1 Introduction

## 1.1 Motivation

A study note is a lossy derivative of a source text, and the lossy steps that produce it are rarely recorded. When such a note is later mined for examination practice, three questions become simultaneously live: does the note state what the source states, is the practice material answerable against the note at all, and can the score attached to the attempt be trusted? The third question is the one most often skipped. A score is ordinarily reported as a number, and a number launders whatever corruption sat upstream of it.

The case reported here is a compact instance of precisely that failure mode. An answer key was supplied with an examination set. The key was internally contradictory — the letter it marked was not the option its own explanation defended — and the contradiction was not subtle. It was detectable by reading the key against itself, without reference to any subject-matter expertise. Despite that, a score had already been produced from it. The purpose of this paper is to document the detection, the reconstruction, the rescoring, and the residue that reconstruction could not remove.

## 1.2 Research questions

- RQ2. What structural and factual defects does a line-level audit of the Module 7 note expose, and are they mutually independent?
- RQ2. Can a supplied answer key be validated for internal consistency, and what is the recoverable structure of a key that fails that validation?
- RQ2. What accuracy does the analyst achieve against the reconstructed key under each of three scoring weightings, and do the figures reconcile?
- RQ2. How is that accuracy distributed across item provenance, and does it track the note’s explicitness on the distinction each item actually tests?

## 1.3 Contributions

The contribution is not a general claim about study materials. It is a worked record of one note, one item set, one defective key, and a reconstruction, reported with the unresolved threads left visible. Two findings survive the reconstruction and are stated as open rather than closed: the defects in the note itself, and the cause of the letter displacement.

# 2 Materials and Method

## 2.1 Artifact localization

The session opened with a request to locate a CompTIA study book within the personal Obsidian vault. A case-insensitive recursive search was executed over the vault. The search returned exactly one matching title.

A single title is a useful result in itself. The search space of a long-lived personal vault is not small, and a duplicate-across-formats situation (one PDF, fifteen derived notes) is a common failure

Table 1: Properties of the located artifact.

| Property                  | Value                                              |
|---------------------------|----------------------------------------------------|
| Matching titles returned  | 1                                                  |
| Author / exam code        | Mark Ciampa; SY0-701                               |
| Stored representations    | Source PDF <i>and</i> directory of extracted notes |
| Number of extracted notes | 15 (one per module)                                |
| Module selected           | Module 7, <i>Identity and Access Management</i>    |
| Length of selected note   | 471 lines, read in full                            |

mode for citation: a reader may cite the PDF for a claim that only the derived note makes. The protocol below therefore requires that every scored answer be anchored to a line in the *module note*, not to the PDF, so that the artifact actually being tested is the artifact being scored.

## 2.2 Scope selection

The user narrowed scope to Module 7. The note is organized around three stated objectives, which partition the item space and give the assessment a defensible coverage target.

Table 2: The three objectives of the Module 7 note.

| #  | Objective                       | Assessed by items on                        |
|----|---------------------------------|---------------------------------------------|
| O1 | Authentication credential types | Knowledge factors; credential modalities    |
| O2 | Authentication best practices   | Password hygiene; breach-response procedure |
| O3 | Access control schemes          | Mandatory, discretionary, role-based        |

## 2.3 Note audit

A line-level audit was performed on the 471-line note prior to item administration. Its purpose was to establish whether the note is a sound basis for scoring at all — a precondition for RQ2 and RQ3, since an unsound basis invalidates any score computed against it. The audit found three defects, recorded in Table 3 and preserved verbatim as findings.

Table 3: Defects surfaced by the line-level audit.

| ID | Type               | Description                                                                                                                                                               |
|----|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1 | Contradiction      | The note states the prevailing industry practice on password expiry <i>and</i> separately issues an explicit recommendation against expiry. A reader cannot satisfy both. |
| D2 | Malformed item     | A review question presents two answer options that are textually identical, making the item unanswerable as written.                                                      |
| D3 | Unsatisfiable item | An item demands two selections, but only one is supportable from the source. The second required selection has no referent in the note.                                   |

The three defects are of different kinds and should not be treated as one class. D1 is a *content* defect: the note is internally inconsistent about a fact, so a reader who studies it may learn either position depending on which passage they encounter. D2 is a *formatting* defect: the item is well-formed in intent but has a degenerate option list, so a correct reader and an incorrect reader are indistinguishable. D3 is a *coverage* defect: the note is silent on a distinction the item tests, so the item is unanswerable from the assigned material regardless of reader expertise. This distinction matters in §4, where D3 resurfaces as the mechanism behind the observed failure cluster.

No correction to the note is reported in the session record, and none is claimed here. Whether D1–D3 originated in the extraction of the PDF into per-module notes, or were present in the source textbook, is not established by the record.

## 2.4 Item set and scoring protocol

Twenty-five examination items were administered. Each item carried a nominal value of four points, giving a nominal total of  $25 \times 4 = 100$  points. The protocol imposed two requirements on the analyst:

- P1.** every item is scored — no item may be skipped, including items the analyst believes are defective;
- P2.** every scored answer carries a line-level citation into the module note.

Requirement P1 is what makes the audit findings actionable: an unscorable item must be scored as incorrect or partial and flagged, not silently dropped. Requirement P2 makes the artifact, rather than the analyst’s recall, the authority for correctness — and it is precisely this property that later allowed the note’s silences to be localized (§3.6).

## 2.5 Key alignment procedure

An answer key was supplied alongside the item set. It failed an internal-consistency check applied before any scoring was attempted. The check requires no subject-matter knowledge: for each entry, the option letter that the key marks as correct must be the option that the key’s own explanatory sentence defends. Two failure modes were observed.

- F1.** *Letter–rationale contradiction.* In roughly three quarters of the key’s entries, the marked letter and the written rationale pointed at different options.
- F2.** *Topical orphaning.* Several entries were topically unrelated to any supplied item — they could not be assigned to a question at all.

The recovery procedure was a subject-matter re-mapping: for each entry, the rationale was read on its own terms and matched to the option it actually defends, independent of the letter printed beside it. The outcome was asymmetric and informative — the rationales were reliable, the letters were not. The key therefore possessed a correct layer (prose) and a displaced layer (labels), which is recoverable in principle: re-deriving the letter from the rationale reconstructs the key.

To test whether the displacement was a property of the key or an artifact of the item set’s option ordering, the user reissued the same twenty-five items with all options reordered within each question. Against the reordered set, the key was internally consistent throughout. This confirms displacement *relative to a particular option ordering*; it does not establish the mechanism that produced the displacement. No further cause is identified in the record, and none is asserted here.

## 2.6 Scoring schemes

Three accuracy figures were produced, under two weighting schemes. They are defined before reporting so that each is independently checkable.

### **W1 — Strict, all-or-nothing.**

Every item is either correct or not. An item answered with a correct but incomplete set of selections is recorded as *partial* and earns no credit. Denominator is all 25 items.

### **W2 — Selection-weighted.**

Single-select items retain W1 treatment. Each multi-select item is credited proportionally to the fraction of its required selections that were correctly made, so an item requiring  $k$  selections and earning  $m \leq k$  correct selections contributes  $4m/k$  of its 4 points.

### **W3 — Strict, restricted.**

W1 applied to the 21 items remaining after excluding the four items judged defective or disputed by either the source note or the key. Denominator is 21.

W3 is reported because the exclusions are not neutral with respect to difficulty: three of them correspond to the audit defects of Table 3, and the fourth is disputed by the key. W1 and W3 are therefore bounds of a kind — W1 includes material the study artifact cannot support, W3 excludes it — and the gap between them is an information-content quantity, not a correction.

## 3 Results

### 3.1 Audit outcome

The audit produced exactly the three defects of Table 3 and no others. No defect of type D2 (degenerate option list) or D3 (unsupportable selection) was found among the twenty-five administered items; the D2 and D3 instances are defects of the note’s own review questions, not of the assessment items. The administered items are, in this respect, cleaner than the source they were drawn from.

### 3.2 Key alignment outcome

Both failure modes F1 and F2 were confirmed and then repaired by re-derivation. The reconstructed key — the rationales with letters recomputed from them — was used as the sole scoring authority for all figures in §3.3. The residual, and it is a real one, is that the cause of the displacement remains unestablished. The option-reordering test narrows the hypothesis space without closing it: it demonstrates that the rationales index the correct *content*, and that the letters indexed a *particular ordering* that the reissued set did not reproduce. It does not identify the process that attached those letters.

### 3.3 Aggregate accuracy under three weightings

Table 5 reports the three figures. All three are derived from a single underlying tally and are therefore mutually constraining; the arithmetic is recomputed in §A.

Three reconciliations hold across the three schemes, and their consistency is the strongest evidence that the figures describe one underlying response pattern rather than three independent tallies.

Table 4: Validation of the supplied answer key.

| Check                           | Outcome               | Interpretation                               |
|---------------------------------|-----------------------|----------------------------------------------|
| Letter matches rationale        | Fail                  | Present in roughly three quarters of entries |
| Entry topically matches an item | Fail                  | Several entries unassignable                 |
| Subject-matter re-mapping       | Rationales valid      | Content layer intact; label layer not valid  |
| Reissue with options reordered  | Consistent throughout | Displacement confirmed; cause unresolved     |

Table 5: Accuracy against the aligned key under three weightings.

| Scheme                | Correct | Denominator | Accuracy              | Note                               |
|-----------------------|---------|-------------|-----------------------|------------------------------------|
| W1 Strict             | 18      | 25          | 72.0%                 | 4 incorrect, 3 partial (no credit) |
| W2 Selection-weighted | —       | 25          | 81.0%                 | 20.25 points equivalent            |
| W3 Strict, restricted | 18      | 21          | $85.7\% \approx 86\%$ | 4 items excluded                   |

- $18/25 = 72.0\%$ , exact.
- $18/21 = 85.7\%$ , which rounds to the reported 86% under W3. This requires that all four excluded items fell among the seven non-correct items; that is consistent with the reported figure, and it is the only allocation that yields it.
- $20.25/25 = 81.0\%$  under W2, implying 2.25 fractional points distributed across the three partial items. The per-item fractions were not retained in the session record, so the *distribution* of that credit is not recoverable here — only its total.

The spread from 72% to 86% is fourteen points on a 100-point nominal scale, arising entirely from scoring policy and item eligibility rather than from any change in the analyst’s responses. This is the practical hazard the case illustrates: the same response set yields three defensible accuracies, and the choice among them is not neutral reporting.

### 3.4 Decomposition by item provenance

The reissued set contained the same twenty-five items with reordered options, of which five were newly introduced. This is an accidental but usable natural split. The five new items were answered before the analyst had encountered them; the twenty carried-over items were items on which the analyst had already anchored during the earlier attempt.

Table 6: Strict accuracy split by item provenance.

| Provenance                           | Items | Correct | Accuracy |
|--------------------------------------|-------|---------|----------|
| Newly introduced (reissued set only) | 5     | 5       | 100.0%   |
| Carried over from earlier set        | 20    | 13      | 65.0%    |
| All items                            | 25    | 18      | 72.0%    |

The split reconciles exactly:  $5 + 13 = 18$ , the W1 correct count. The thirty-five-point gap

between the two strata is the largest single effect in the study, and it runs in a direction that must be interpreted with care. It is *not* evidence that the new items were easier in the abstract. The five new items are, by the report, drawn from the same note and the same three objectives.

### 3.5 What the five new items test

The five items answered without exception cover, in the order reported: a response to a suspected multi-factor fatigue attack; digest comparison as protection of a banking credential; the coexistence of mandatory integrity control with discretionary access control; justification logging together with session revocation for privileged access; and the access control scheme whose characteristic weakness is that executed programs inherit the account holder's privileges.

Read against the failure cluster in the next subsection, this list is diagnostic. Each of the five tests a *relational* or *procedural* property — what to do when an attack is suspected, what a digest comparison defends, which controls may co-exist, what a privileged session must log and revoke, what a weakness consists of. Each is answerable from a note that states the relevant principle, whether or not the note enumerates it as a review topic.

### 3.6 Localization of failures

The degradation among the carried-over items was not uniform. It was concentrated in items where the source note is *silent* on the distinction the item tests. Three such items are named explicitly in the record:

- S1.** precomputed hash databases — a distinction the note does not draw;
- S2.** the goal of *using* rather than creating a password mask — a distinction the note does not draw;
- S3.** whether a hardware security key transmits authentication codes — a distinction the note does not draw.

These are all instances of defect class D3. The item is well-formed, the option set is well-formed, and the key is aligned; the failure is that the assigned artifact contains no proposition from which the answer can be derived. An examinee who knows the subject matter and an examinee who has read the note are in the same position, which is to say no position. This is the mechanism that reconciles the two strata: where the note is explicit, accuracy is near total, and where the note is silent, the note cannot distinguish knowledge from recall of nothing.

Crucially, the two success strata are not disjoint from the two failure strata by provenance. Carried-over items that test propositions the note states explicitly were answered correctly. New items that test relations the note states were answered correctly. What predicts failure is silence, not novelty — although novelty and exposure are themselves confounded in this design, as discussed next.

## 4 Discussion

### 4.1 The note is sufficient, not complete

The central substantive finding is that Module 7 performs as a *sufficient* but not a *complete* basis for assessment. Sufficiency here means that for a proposition the note states, a reader who studies

the note can answer correctly — the correlation between explicitness and accuracy is strong enough that failure is rare. Completeness would mean that every proposition an item can test is present in the note; it is not, and the three S-class failures are the direct evidence.

This has a practical consequence for extraction pipelines. A note derived from a textbook by summarization inherits the textbook’s coverage only to the extent that the summarizer preserved it. Coverage loss is silent, because a reader cannot detect the absence of a proposition — only its absence of benefit. D3 is the observable residue of that loss, and it would not have been visible in this study without the line-level citation requirement P2.

## 4.2 Contamination and its direction

The thirty-five-point gap in Table 6 cannot be read as a clean measure of item difficulty. The five new items were administered to an analyst who had just completed a full pass of the same note, and who had just been alerted — by the audit — to the note’s failure modes. The twenty carried-over items were answered after that analyst had already anchored on them during the earlier attempt. Novelty, exposure count, and prior commitment to a specific option all separate these strata simultaneously, and the design does not permit them to be separated.

The honest reading is therefore directional rather than quantitative: the new items are consistent with a study effect, and the carried-over items with a commitment effect, and this study cannot apportion between them. What can be said is that neither stratum produced the S-class failure pattern — the silent-distinction failures occurred among items whose content the note never supplied, which is a fact about the artifact and not about the analyst’s state.

## 4.3 Open findings

Two threads survive the reconstruction and are recorded here as open rather than closed.

- **The note’s defects are unrepaired.** D1, D2, and D3 were identified; no correction is recorded. D1 in particular is a live hazard: a reader who meets the industry-practice statement and the anti-expiry recommendation in either order may learn a different answer depending on which they retain.
- **The letter displacement has no established cause.** The reordering test confirmed displacement and localized it to option ordering. Whether the letters were carried over from a different ordering of the same items, transcribed from a key built for a different set, or corrupted by a mechanical pass, is not determined by the record. No mechanism is proposed here.

## 4.4 Implications for assessment practice

Three transferable practices follow, each of a different scope.

1. *Validate a key against itself before using it.* The letter–rationale check requires no subject-matter expertise and detected a corruption that had already produced a score. It should precede, not follow, scoring.
2. *Score defective items, then report both denominators.* Protocol P1 combined with W1 and W3 makes the cost of artifact quality visible as a fourteen-point spread instead of hiding it inside a single number.

3. *Cite the artifact, not recall.* The line-level citation requirement is what made the S-class localization possible. Without it, those seven non-correct items would have been indistinguishable from ordinary errors.

## 5 Conclusion

A single CompTIA Security+ module note was assessed end to end. The note (471 lines, three objectives, drawn from a vault holding one PDF and fifteen derived module notes) contains three defects spanning content, formatting, and coverage. A 25-item, 100-point protocol with mandatory line-level citation was administered against it. The supplied answer key was internally inconsistent in roughly three quarters of its entries and was reconstructed by subject-matter re-mapping of its rationales; reissue of the items under a new option ordering confirmed the displacement while leaving its cause open.

Against the aligned key, accuracy was 72% under strict all-or-nothing scoring, 81% under selection-weighted scoring, and 86% under strict scoring with defective and disputed items excluded — three figures reconciling to a single tally. Decomposition showed 5/5 on newly introduced items against 13/20 on carried-over items, with failure concentrated precisely where the note is silent on the tested distinction and near-absent where it is explicit.

The generalizable claim is narrow and worth stating exactly: an extracted study note is a sufficient but incomplete assessment substrate, and a score computed against one is a statement about the note’s coverage unless the score is anchored to it line by line. The note’s three defects and the cause of the key’s letter displacement remain open.

## References

- [1] Ciampa, M. *CompTIA Security+ Study Guide* (exam code SY0-701). Retained locally as a source PDF and as a directory of fifteen extracted per-module notes within the author’s Obsidian vault.
- [2] *Module 7: Identity and Access Management (SY0-701)*, 471-line extracted study note, local artifact. not recorded in the session summary; recover before archiving.
- [3] *Examination item set*, 25 items at 4 points each, analyst-supplied.
- [4] *Purported answer key*, analyst-supplied, subsequently shown to be letter-displaced.

*No external literature was consulted. The four references above are the internal artifacts of the study and are the complete reference set.*

## A Arithmetic Checks

All figures reported in §3.3 and §3.4 are recomputable from the counts stated in the session record. The checks below use no information beyond those counts.

$$\begin{aligned}
&\text{Nominal total} = 25 \times 4 = 100 \text{ points} \\
&\text{W1 (strict)} = \frac{18}{25} = 72.0\% \\
&\text{W2 (sel.weighted)} = \frac{18 + 2.25}{25} = \frac{20.25}{25} = 81.0\% \\
&\text{W3 (strict, excl.)} = \frac{18}{25 - 4} = \frac{18}{21} = 85.714 \dots \% \approx 86\% \\
&\text{Decomposition} = \frac{5}{5} = 100.0\% \qquad \frac{13}{20} = 65.0\% \\
&\text{Reconciliation} = 5 + 13 = 18 \quad \checkmark
\end{aligned}$$

The W2 residual of 2.25 points over three partial items is consistent as a total; the per-item fractions behind it were not retained and are not reconstructed here. The W3 figure is reproduced only under the allocation in which all four excluded items fall among the seven non-correct items, which is the unique allocation yielding the reported value.

## B Open Findings Register

Table 7: Unresolved items carried forward from the session.

| ID | Status | Statement                                                                                                                 |
|----|--------|---------------------------------------------------------------------------------------------------------------------------|
| D1 | Open   | Note states industry password-expiry practice and separately recommends against expiry; contradiction not repaired.       |
| D2 | Open   | Note contains a review question with two textually identical options; not repaired.                                       |
| D3 | Open   | Note contains an item requiring two selections with only one supportable; not repaired.                                   |
| K1 | Open   | Exact count of letter-rationale contradictions not retained; reported only as roughly three quarters of entries.          |
| K2 | Open   | Cause of letter displacement not established; option-reordering test confirms displacement without identifying mechanism. |
| W2 | Open   | Per-item fractional credit behind the 81% figure not retained; only the 2.25-point total is recoverable.                  |

**Provenance of this document.** Compiled from a distilled session summary by the `latex-paper-writer` subagent. Every count, topic, and topic-level claim in the body is grounded in that summary. Inferred structural readings — the objective-to-failure-cluster mapping in Table 2 and the “relational/procedural” characterization in §3.4 — are labelled as interpretation. No external source was read, and no empirical result beyond the session’s own tallies is asserted.